

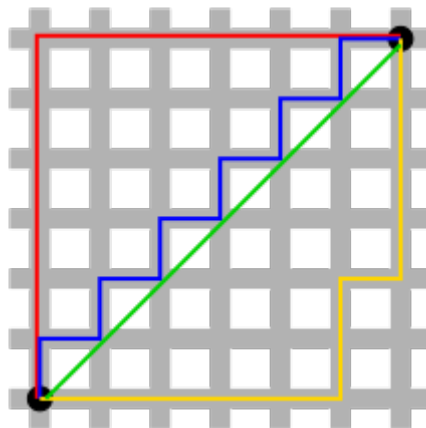
Manhattan Distance

Definition

For points p and q on a plane, we can define the distance between them as the sum of the differences between their x and y coordinates:

$$d(p,q) = |x_p - x_q| + |y_p - y_q|$$

Defined this way, the distance corresponds to the so-called [Manhattan \(taxicab\) geometry](#), in which the points are considered intersections in a well designed city, like Manhattan, where you can only move on the streets horizontally or vertically, as shown in the image below:



This images show some of the smallest paths from one black point to the other, all of them with length 12 .

There are some interesting tricks and algorithms that can be done with this distance, and we will show some of them here.

Farthest pair of points in Manhattan distance

Given n points P , we want to find the pair of points p,q that are farther apart, that is, maximize $|x_p - x_q| + |y_p - y_q|$.

Let's think first in one dimension, so $y=0$. The main observation is that we can bruteforce if $|x_p - x_q|$ is equal to $x_p - x_q$ or $-x_p + x_q$, because if we "miss the sign" of the absolute value, we will get only a smaller value, so it can't affect the answer. More formally, it holds that: